

The Spinning Brain Project

Complete Status, Architecture, and Results (v2 — all stages)

Consolidated working document

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One-line status. A novel “spinning” (phase-based) reasoning substrate now demonstrates — at *small/toy scale*, all learned with *no hard-coded values* — **every stage of the plan**: it reasons causally, stays cheap, holds many skills without forgetting, is multi-modal (vision + audio + symbols with fusion), knows what it doesn’t know, refuses to fabricate, **seeks missing knowledge by asking in language**, generates language, and runs an **exploration→continuous-learning loop** (curiosity-driven discovery with online learning and zero forgetting — the “alive” loop). The one hard limitation found (deep exact modular arithmetic on the bare substrate) was *defeated* with a small add-on. **12 capabilities verified by experiment; 10 are in the integrated suite** (`spinning_brain.py`, **10/10 PASS**) and **Stage 5/6 have standalone verified scripts**. One honest open strain: *scaled* compositional generalization. This demonstrates *mechanisms*, not a finished product.

1 The Goal

Build a new kind of AI “brain” that is, all at once: **(1) Cheap**, **(2) Excellent at reasoning** (thinks like a brain, not a fluent-text generator), **(3) Omni** (text, images, audio), **(4) Alive** (continuous learning, no forgetting). Expanded: a “conscious-like” reasoner — thinks then answers; if it doesn’t know, *seeks*; never fabricates; no hard-coded values.

Honest translation of “conscious-like.” “Conscious” is *not* adopted literally (undefined, untestable). It is translated to the testable cluster underneath: **self-awareness about its own knowledge** — models its uncertainty, knows its knowledge boundary, acts to reduce uncertainty (seeks), refuses to fabricate. “Never wrong” → “doesn’t confidently guess when it doesn’t know.”

2 The Architecture

Core recurrent step, iterated T times:

$$F(h, x) = (1 - \eta)h + \eta \tanh(Wh + Ux + b), \quad W = -\rho(QQ^\top) + S,$$

Q orthonormal (QR of a learned matrix), S skew-symmetric $\Rightarrow$ **rotational dynamics** (orbits, not fixed points). **Validated discovery:** the trained model *reasons by spinning, not settling* — the answer lives in the *phase* of a non-converging oscillation, proven causal by the brain-swap test (swapping the mid-thought flips the answer 100%; random-state control scatters to chance). Four supporting theorems (halting normalization; $E[N] \leq 1/\varepsilon^2$ compute bound; exact-gradient training; a contraction condition the trained model provably ignores — recorded honestly).

3 All Verified Capabilities

Capability	Result (measured)	Status
Reasoning is causal [Part 2]	parity 1.00; brain-swap flip 1.00; random control 0.49	[DONE]
Consolidation [Parts 1+2+4]	one ~45K-param skill-conditioned core: parity-3 1.00, sum-of-3 1.00, max 1.00, no forgetting	[DONE]
P5 knows-what-it-doesn't-know	confidence 1.00 solvable vs 0.23 unanswerable (gap 0.76)	[DONE]
P6 refuses to fabricate	answers known 100% (0% needless abstain); abstains 96% on unknowable	[DONE]
P7 seeks (toy)	non-memorizable facts: answer-with 1.00; without 0.20 (=chance)	[DONE]
P8 generation	autoregressive, free-running: token 1.00, whole 1.00, no drift	[DONE]
Ceiling fix	clean-state recurrence: mod-5 sum at length 8 and 16 = 1.00, scales (~1K params)	[DONE]
Omni [Part 3]	vision 1.00, audio 1.00, audio+vision fusion 1.00 (control-proven binding)	[DONE]
Language: parse	compositional generalization, held-out novel 1.00	[DONE]
Language: generate	meaning→sentence, free-running, held-out 1.00, correct EOS, no drift	[DONE]
Stage 5: language seeking	generates queries <i>in words</i> (“value of entX”, 100% exact); answer 1.00 with the loop, 0.13 with random values; facts randomized/episode	[DONE]
Stage 6: alive loop	curiosity (own-uncertainty) discovers all facts in 9.3 probes vs 20.2 random; online learning; final 1.00; first-vs-last retention gap 0.00	[DONE]

4 The One Hard Limit — Found and Defeated

The bare spin cannot learn *many-wrap exact* modular arithmetic (rapid oscillation $\Rightarrow$ hostile loss landscape; three principled fixes failed; the atomic op $(r+d) \bmod 5$ *is* learnable at 100%). True cause: **state drift** carrying a running count in continuous state. **Defeated** by the clean-state recurrence (commit to a discrete value each step, straight-through, feed back): exact accumulation at lengths 8 and 16, scaling, ~1K params. Honest caveat: an add-on module beside the bare spin — exactly the predicted “discrete counter.”

5 Honest Scope and Open Boundaries

- **Toy scale throughout** (vocab ≤ 35 words, sentences ≤ 11 tokens, 8×8 grids, synthetic tones). Mechanisms, not a product.
- **The scale strain (the honest open problem)**. At 35 words + recursive grammar (depth 1–3), train whole-parse is 1.00 but **held-out is 0.65** (slots 0.85–1.00; climbing slowly with budget; chance ≈ 0.0008). First real train/held gap: scaled compositional generalization is *partial*. The frontier is genuinely scale.
- **Relational vision partial** (~ 0.76 on “A above B”); direct readouts (localization, aggregation) are 1.00. Theme: direct-readout-strong, multi-element-comparison-weak.
- **Bare-spin arithmetic limit** stands; use the clean-state add-on for exact counting.

- **Record corrections kept visible:** early “sum-of-6” claims did not reproduce under clean isolation and are treated as unverified; the table above contains only re-verified numbers.

6 Method and Discipline

Every claim has a checkable number; wrong hypotheses are killed openly (e.g. “spinning resists forgetting” — false; four consolidation “fixes” that were a misdiagnosis of an unlearnable task). The governing rule, earned the hard way: **isolate before fixing** — a result constant across many interventions means you are solving the wrong problem. It caught the ceiling misdiagnosis, a generation indexing bug (0.00 *on training* $\Rightarrow$ bug, not learning failure), and an infinite dataset loop (asked 400 unique sentences of a 336-max grammar).

7 Roadmap — All Stages

[DONE] **Stage 1** Parts 1+2+4 consolidated (cheap, causal reasoning, no forgetting, one core).

[DONE] **Stage 2** Omni: symbols + vision + audio, control-proven fusion.

[DONE] **Ceiling** diagnosed (state drift) and defeated (clean-state recurrence).

[DONE] **Stage 3** Language understanding: composition, grammar/order, long-range (30 distractors, no fade).

[DONE] **Stage 4** Language generation: meaning $\rightarrow$ sentence, compositional, no drift.

[IN PROGRESS] **Scale-up** the open frontier; first strain measured and recorded (held-out 0.65 at 35-word recursive grammar).

[DONE] **Stage 5** Seeking at language scale: asks *in words* for exactly what it lacks; answers because of what it fetched; nothing memorizable.

[DONE] **Stage 6** Exploration $\rightarrow$ continuous-learning: curiosity from its own uncertainty ($2\times$ more efficient than random), online learning, zero forgetting — the “alive” loop closed.

8 The Code

- `spinning_brain.py` — consolidated substrate + 10 capabilities as an integrated test suite (latest run **10/10 PASS**). `python spinning_brain.py` or `python spinning_brain.py <name>`.
- `stage5_langseek.py` — language-scale seeking (verified: query-language 1.00, answer 1.00, random-control 0.13).
- `stage6_explore.py` — the alive loop (verified: 9.3 vs 20.2 probes, retention gap 0.00).
- `scale_language.py` — the scale-step that measured the held-out strain (0.65).

Bottom line. All six stages of the plan now have working, verified demonstrations — including the two endgame capabilities: *asking in language for what it lacks* and the *curiosity-driven, non-forgetting learning loop*. One honest open strain (scaled compositional generalization) marks the real frontier: **scale**. The mechanisms of the goal — a cheap, causally-reasoning, omni, alive, self-knowing brain — are demonstrated and reproducible.